NATIONAL UNIVERSITY OF SINGAPORE

EXAMINATION FOR

(Semester I: 2019/2020)

EE5507 – ANALOG INTEGRATED CIRCUIT DESIGN

Nov / Dec 2019 - Time Allowed: 2.5 Hours

INSTRUCTIONS TO CANDIDATES:

1. This paper contains **FOUR** (4) questions and comprises **SEVEN** (7) printed pages.
2. Answer all **FOUR** (4) questions.
3. Each question carries 25 marks.
4. This is an OPEN BOOK examination.
5. Calculators can be used in the examination.
6. You may assume the following constant:
   $q=1.6021765\times10^{-19}\ C \quad k=1.38064852\times10^{-23}\ m^2\ kg\ s^{-2}\ K^{-1}$

Q.1

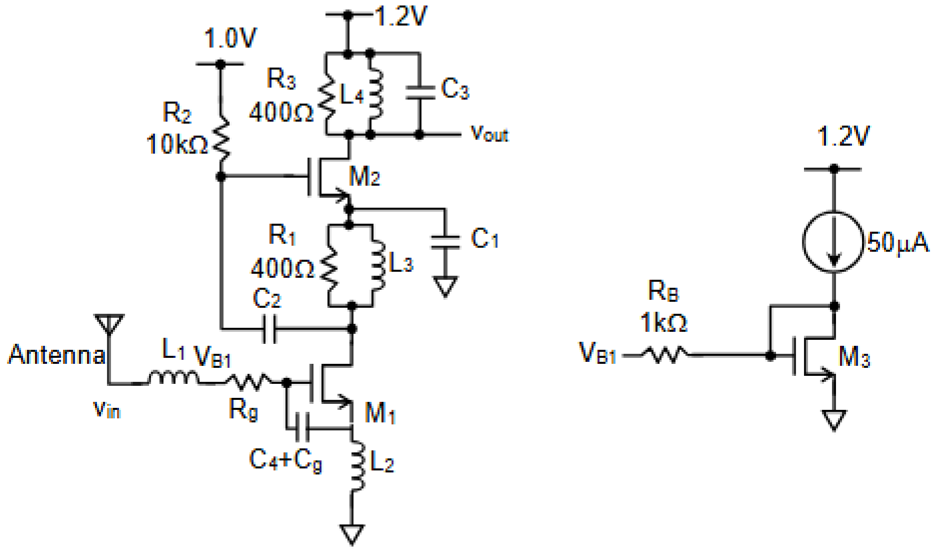

Fig. Q1

A current reuse two stage low noise amplifier (LNA) is shown in Fig. Q1. Assume all the NMOS body are connected to their source respectively. The supply voltage is 1.2V. The device parameters are given as follows:

(Hint: You might want to consider inductor as short circuit during DC analysis and open circuit during AC analysis. This is 2-stage amplifier, i.e. CS with inductor degeneration followed by CS amplifier, where L_1 and L_2 can be chosen for impedance matching and frequency tuning.)

- V_{THN}=0.238V, $\mu_n C_{ox}$=386μA/V^2, λ_n=0.8V^{-1}, C_{ox}=15fF/μm^2, R_{sheet_gate}=15Ω/□.

a) Determine W for M_1 such that C_{GS}=50fF. Also determine the number of fingers needed such that R_g<10Ω. (You may assume that C_{GS} is about 2/3 of the total M_1 oxide capacitance, and L is 0.13μm. R_g is related to R_{sheet_gate}.)

(4 marks)

b) Determine L_1 and L_2 such that you have matching impedance of 50Ω at 2.4GHz. You may assume $g_{m,M1}$=25mA/V, R_g=10Ω and C_4+C_{gs}=800fF.

(4 marks)

c) Design the transistor M_2 and M_3 so that the overall voltage gain (V_{out}/V_{in}) is 40. The biasing V_{B1} is generated such that M_1 has $g_{m,M1}$ of 25mA/V.

(8 marks)

d) Determine $V_{S,M2}$ and thus the minimum allowable output voltage, i.e. $V_{out,min}$.

(4 marks)

e) Determine appropriate value for C_1 and C_2. Justify your choice with reasoning.

(5 marks)

Q.2

a) Fig. Q2(a) shows the single-ended switched-cap opamp with offset cancellation.

i) Draw the corresponding differential switched-cap opamp circuit that has offset cancellation capability, i.e. the circuit should take in differential $V_{in}(V_{inp}-V_{inn})$ and give differential $V_{out}(V_{outp}-V_{outn})$. (Hints: You should use differential opamp. Try to mirror the circuit for the other side, and connect all the ground point to V_{CM}, where V_{CM} is the DC common mode voltage.)

(5 marks)

ii) Based on (i), derive the charges stored in C and C_S during ϕ_1 and ϕ_2, and show that V_{OS} does not appear at V_{out}.

(5 marks)

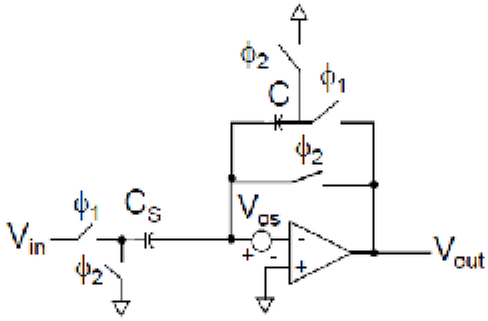

Fig. Q2(a)

b)

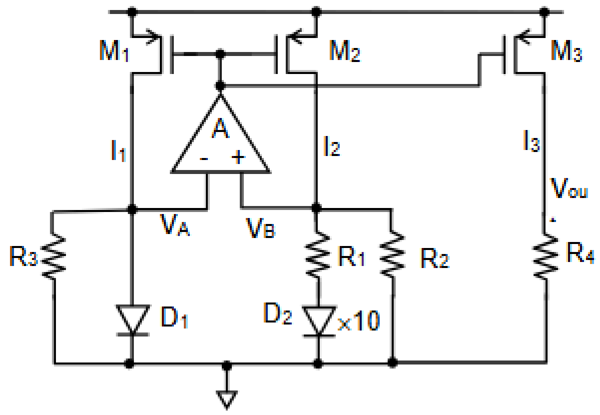

Fig. Q2(b)

For the circuit in Fig. Q2(b), assume that $I_{S,D2}=10\times I_{S,D1}$, $R_2=R_3$, and M_1, M_2 and M_3 are identical.

i) Show that I_{R1} is proportional to absolute temperature, i.e. PTAT.

(5 marks)

ii) Show that I_{R2} is proportional to V_{D1}, and thus V_{out} can be made temperature independent. (Hints: V_{D1} is similar to V_{BE} of BJT covered in lecture.)

(5 marks)

iii) Assume $V_{D1}=1.26-0.003\times T$, where T is the absolute temperature in Kelvin, determine the R_1, R_2 and R_4 such that V_{out} is temperature independent. What is the advantage of this circuit compared to conventional bandgap reference?

(5 marks)

Q3.

a) Consider the following filter transfer-function

$$H(s) = \frac{K(a_2 s^2 + a_0)}{b_4 s^4 + b_3 s^3 + b_2 s^2 + b_1 s + b_0}$$

where K=2, a_2=1, a_0=9, b_4=1, b_3=1.3, b_2=1, b_1=0.75, b_0=1

i) Determine the DC gain in dB.

(2 marks)

ii) Calculate the rate of attenuation in dB per decade at high frequencies.

(3 marks)

iii) Determine the frequencies where the attenuation would be infinite.

(5 marks)

b) For the following lowpass filter specifications:

Passband f_p = 1MHz
Stopband f_s = 4MHz
Passband ripple a_{max} = 1dB
Minimum stopband attenuation a_{min} = 50dB

i) Determine the required filter order N and its Magnitude Transfer function.

(7 marks)

ii) Derive the Butterworth Filter transfer function T(s) that meets this specification. Also derive its final form as a product of 1st and/or 2nd order polynomials, which is needed for the filter implementation. What is the value of the frequency scaling factor $\mathbf{k_f}$?

(8 marks)

Q4.

a) Find the total output noise (in dBm) if 2 noise sources of power -60 dBm and -66 dBm are summed together.

(3 marks)

b) A basic track and hold circuit is shown in Fig. Q4(b). Assume that the input signal Vi has a full-scale range from 0 to 1V.

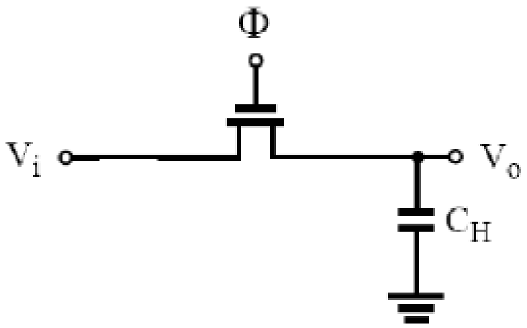

Fig. Q4(b)

i) If several track-and-hold circuits of this type are used in a 10-bit ADC and the total input-referred thermal noise power is found to be $10 \cdot kT/C_H$, determine the capacitance (C_H) needed such that the total input-referred thermal noise rms voltage is equal to 0.25×LSB when T = 27°C.

(2 marks)

ii) Compared to the case where only quantization noise is present, how much reduction (in dB) in dynamic range (DR) is due to the input-referred thermal noise rms voltage above?

[**Hint**: For quantization noise power (E_Q), E_Q= $\mathbf{V_{LSB}^2/12}$, and DR=(signal power)/(noise power)]

(5 marks)

c) For the transimpedance amplifier (TIA) circuit shown in Fig. Q4(c) at T = 27°C,

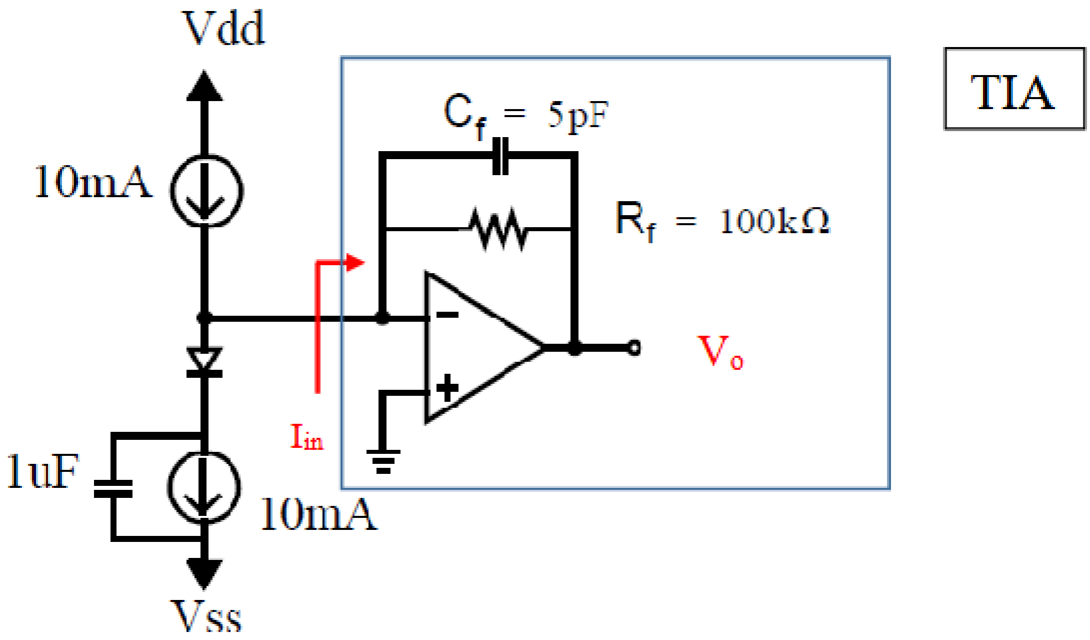

Fig. Q4(c)

i) Find the total output noise ($V_{o,noise}$) for the circuit in Fig. Q4(c). You may assume that the opamp and current sources are ideal and noiseless.

(7 marks)

ii) Find the total output noise in the case where the opamp is non-ideal with gain transfer function of

$$A(s) = \frac{A_0}{\left(\frac{s}{\omega_p} + 1\right)} = \frac{\omega_u}{(s + \omega_p)}$$

when ω_u = 20M rad/s. What is the total output noise when ω_u = 200k rad/s? State any assumptions you make regarding A_0, ω_p and ω_u. You may assume the opamp is still noiseless.

(**Hint**: Derive the transfer function of this transimpedance amplifier TIA (V_o/I_{in})

(8 marks)